USB e alimentazione

File: usb_alimentazione.kicad_sch

MCU

File: mcu.kicad_sch

Display (cavo FFC)

File: display.kicad_sch

Periferiche I2C e pulsante

File: periferiche.kicad_sch

Blocchi (vedi docs/02-architettura-hardware.md e docs/12-selezione-componenti.md):
- USB e alimentazione: USB-C 2.0 (CC 5,1 k), PTC 500 mA, ESD USBLC6-2SC6, LDO AMS1117-3.3, punti di test.
- MCU: ESP32-C3-MINI-1-N4, RC su EN, tasti RESET/BOOT, disaccoppiamento.
- Display: connettore FFC 16 poli 0,5 mm verso la schedina portadisplay, driver retroilluminazione PWM, predisposizione touch con resistenze 0 ohm NON montate (DEC-019).
- Periferiche: pull-up I2C, connettori JST-SH 4 poli tipo Qwiic per SHT40 (J3) e modulo RTC DS3231 (J4), pulsante utente remoto su connettore (J5).

Assegnazione GPIO (ESP32-C3-MINI-1):
IO0 TP_INT | IO1 TP_RST | IO2 I2C_SDA | IO3 BTN_USER | IO4 LCD_DC | IO5 LCD_RST | IO6 LCD_SCLK
IO7 LCD_MOSI | IO8 I2C_SCL | IO9 BOOT | IO10 LCD_CS | IO18 USB_D- | IO19 USB_D+ | IO20 (RXD0) LCD_BL
IO21 (TXD0) UART_TX su punto di test. Strapping: IO2 e IO8 con pull-up 4,7 k (I2C), IO9 pull-up 10 k.

Indirizzi I2C: SHT40 0x44, DS3231 0x68 (EEPROM assente sul modulo Adafruit), touch CST816S 0x15 (da verificare).
Componenti DNP (non montati): R11 R12 R13 R14 (touch).

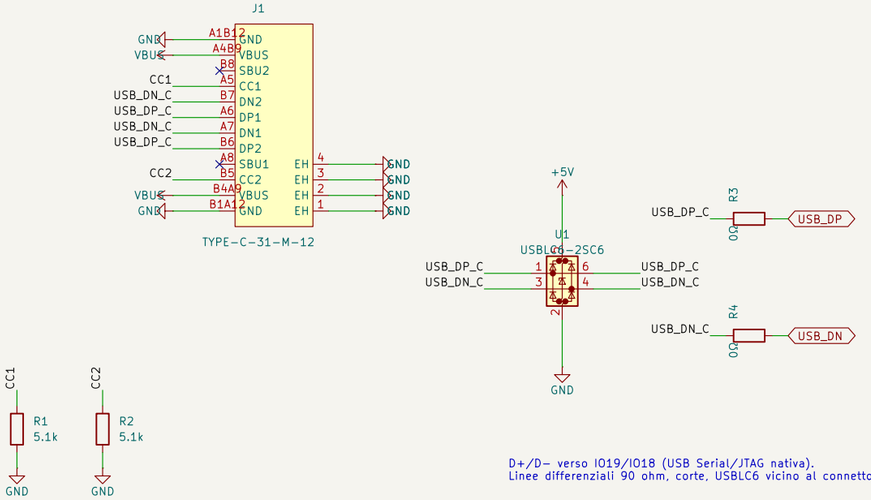
RTC esterno: modulo Adafruit DS3231 STEMMA QT su J4 (DEC-024).
ESP32-C3-MINI-1-N4, USB-C, AMS1117-3.3, FFC 16 poli verso portadisplay.
Schema gerarchico. Rev 0.1 NON validata: revisione visiva del proprietario richiesta.
PoliClock

Sheet: /
File: policlock-main.kicad_sch

Title: PoliClock — scheda principale

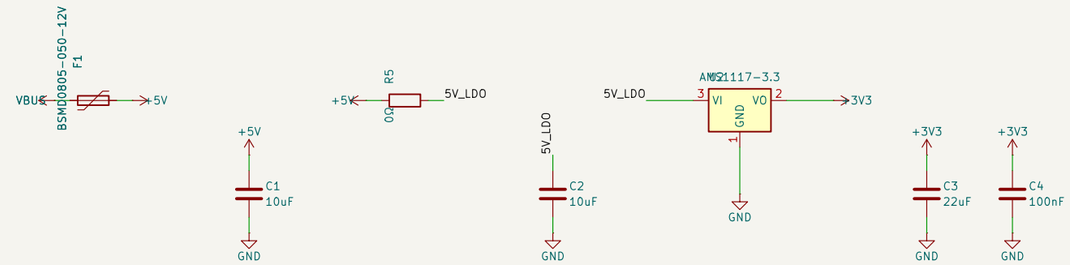
Size: A3	Date: 2026-11-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/5

Ingresso USB-C, CC, ESD



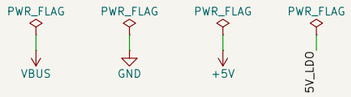
D+/D− verso IO19/IO18 (USB Serial/JTAG nativa).
Linee differenziali 90 ohm, corte, USBLC6 vicino al connettore.

Protezione VBUS e regolatore 3,3 V

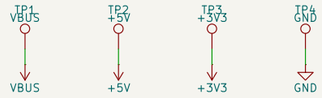


AMS1117: Cout 22 uF (datasheet: 22 uF tantalio per la stabilita').
Verificare stabilita' con MLCC al bring-up; Pd ~ (5-3,3) V x I.

PWR_FLAG

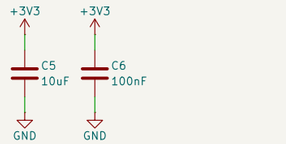


Punti di test

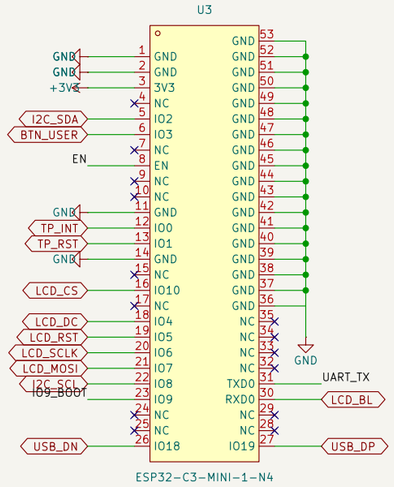


PoliClock		
Sheet: /USB e alimentazione/		
File: usb_alimentazione.kicad_sch		
Title: PoliClock main — USB e alimentazione		
Size: A3	Date: 2026-11-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/5

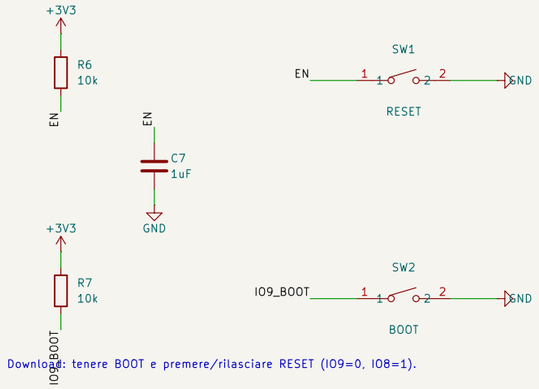
Modulo ESP32-C3-MINI-1-N4 (4 MB flash, antenna PCB: zona di esclusione)



Disaccoppiamento vicino al pin 3V3 del modulo

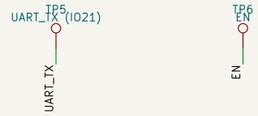


EN (reset) e BOOT



Download: tenere BOOT e premere/rilasciare RESET (IO9=0, IO8=1).

Debug



Log e flash principali via USB nativa;
TXD0 su punto di test per debug di riserva.

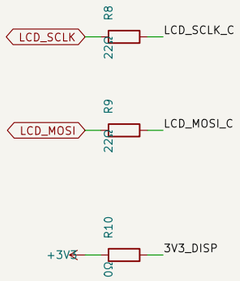
PoliClock

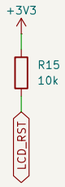
Sheet: /MCU/
File: mcu.kicad_sch

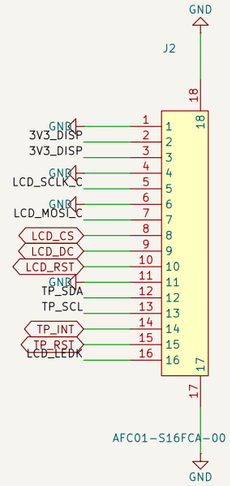
Title: PoliClock main — MCU

Size: A3	Date: 2026-11-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/5

Collegamento alla schedina portadisplay (FFC 16 poli, passo 0,5 mm, contatti in basso)

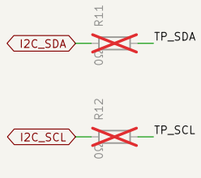


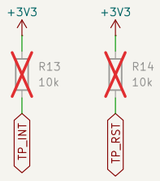




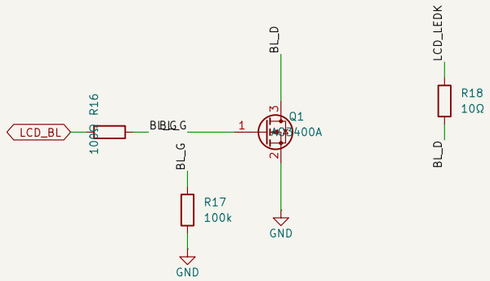
Pinout cavo (identico sulla portadisplay):
1 GND 2 3V3 3 3V3 4 GND 5 SCLK 6 GND 7 MOSI 8 CS
9 DC 10 RST 11 GND 12 TP_SDA 13 TP_SCL 14 TP_INT 15 TP_RST 16 LEDK
Orientamento cavo (tipo A/B) da fissare nel layout: pin n → pin n.

Touch predisposto (DEC-019): R11–R14 NON montati.
Montare R11/R12 per collegare il touch al bus I2C.





Retroilluminazione: LEDA a 3V3 sulla portadisplay, LEDK commutato a massa.
R18 limita la corrente: valore TBD dal datasheet del pannello acquistato.



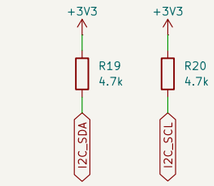
PoliClock

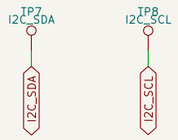
Sheet: /Display (cavo FFC)/
File: display.kicad_sch

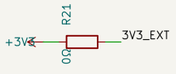
Title: PoliClock main — Display (cavo FFC)

Size: A3	Date: 2026-11-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/5

Bus I2C (SHT40 0x44, RTC DS3231 0x68, touch 0x15)

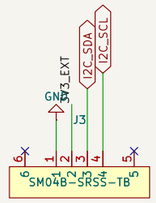


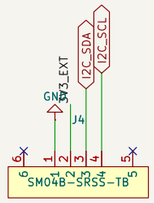




Il modulo Adafruit DS3231 ha gia' pull-up 10 k verso Vin:
in parallelo a R19/R20 -> ~3,2 k, accettabile.

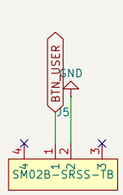
Connettori Qwiic / STEMMA QT (JST-SH 4 poli: 1 GND, 2 3V3, 3 SDA, 4 SCL)

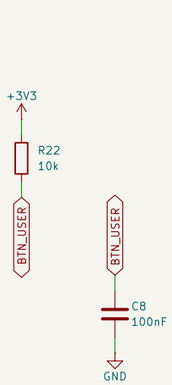




J3: modulo breakout SHT40 commerciale (header 2,54 mm), cavo JST-SH 4 poli -> jumper femmina; ordine dei pin dalla serigrafia del modulo (DEC-028).
J4: modulo Adafruit DS3231 Precision RTC STEMMA QT (5188) con pila CR1220 (DEC-024).

Pulsante utente





IO3: risveglio da deep-sleep possibile (GPIO0-5).
Solo pulsante remoto su J5 (DEC-032): nessun tasto USER su scheda.

PoliClock		
Sheet: /Periferiche I2C e pulsante/		
File: periferiche.kicad_sch		
Title: PoliClock main – Periferiche I2C e pulsante		
Size: A3	Date: 2026-11-07	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/5